

# PAPER\_053: NGC 2264 Cone Nebula Star-Forming Region: 8-Test UQFF Validation of Compressed Gravity, Electromagnetic Dominance, and Star Formation Rate

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**Author:** Daniel T. Murphy

**Framework:** UQFF Star-Magic ( $\kappa = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ )

**Date:** March 7, 2026

**Validator:** validate\_all\_models.py — NGC2264Model: **8/8 PASS** ✓

**Source Module:** CondensedPhysics.py (NGC2264Model), validate\_all\_models.py

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## Abstract

NGC 2264 — the Cone Nebula and Christmas Tree Cluster star-forming complex in Monoceros — is the primary UQFF reference system for active T-Tauri star formation. The UQFF NGC2264Model passes all 8 validation tests, covering the gravitational field, Hubble expansion term, T-Tauri star formation mass, protostellar disk erosion energy, electromagnetic compressed gravity, total compressed gravity, resonance amplitude, and EM dominance ratio. The star-forming environment is characterized by compressed gravity  $g_{\text{compressed}} = 1.0533 \times 10^{-2}$  and resonance amplitude  $R = 1.1586 \times 10^{-2}$ . All 8 tests pass with predicted/observed ratios between 0.9980 and 1.0011.

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## 1. System Identification

NGC 2264 (also designated Caldwell 41) is a young ( $\sim 2$  Myr) open cluster and HII region at a distance of  $\sim 720$  pc in the constellation Monoceros. It contains: - The **Cone Nebula** (dark nebula, photodissociation region) - The **Christmas Tree Cluster** (OB association,  $\sim 100$  members) - Active T-Tauri and pre-main-sequence (PMS) stars - Outflow jets and Herbig-Haro objects

**UQFF Classification:** Active star-forming region, EM-dominated regime ( $a_{\text{EM}}/g_{\text{total}} > 0.99$ )

UQFF parameters for NGC2264: | Parameter | Value | Source | |-----|-----|-----| | Distance |  $\sim 720$  pc | Observed | | Mass (cluster) |  $\sim 500 M_{\odot}$  | Literature | | Star formation rate |  $\sim 1.5 M_{\odot}/\text{Myr}$  | UQFF  $M_{\text{sf}}$  calibration | | Age |  $\sim 2$  Myr | Literature | | Redshift  $z$  |  $\sim 0$  (local) | Distance |

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## 2. The 8 UQFF Tests

### Test 1: Gravitational Field $g_{\text{grav}}$

$$g_{\text{grav}} = G \times M_{\text{cluster}} / r_{\text{eff}}^2$$

- Predicted:  $5.9336 \times 10^{-11} \text{ m/s}^2$
- Expected:  $5.9270 \times 10^{-11} \text{ m/s}^2$
- Ratio: 1.0011 (**PASS**)

This ultra-high agreement (0.11% error) confirms that the UQFF gravitational computation for NGC2264 uses the correct stellar mass and effective radius, and that the standard Newtonian term in the full UQFF expression matches to better than 0.2%.

### Test 2: Hubble Expansion Factor

The UQFF compressed gravity includes a Hubble term for cosmological context:

$$H_{\text{factor}} = 1 + H_0 \times (z + t_{\text{age}}/t_H)$$

For a local system ( $z \approx 0$ , age  $\approx 2 \text{ Myr}$ ): - Predicted: 1.0002 ( $H_0$  correction for local cluster) - Expected: 1.0002 - Ratio: 1.0000 (**PASS**)

The essentially unity Hubble factor confirms NGC2264 is unaffected by cosmological expansion — the cluster is fully bound and local. The 0.02% Hubble term is the proper cosmological correction at 720 pc.

### Test 3: T-Tauri Star Formation Mass $M_{\text{sf}}$

The UQFF  $M_{\text{sf}}$  term computes the expected mass of T-Tauri stars currently forming in the cloud:

$$M_{\text{sf}} = \Sigma_{\text{gas}} \times \epsilon_{\text{ff}} \times t_{\text{ff}}$$

where  $\epsilon_{\text{ff}}$  is the UQFF star formation efficiency per free-fall time, calibrated from the [SCm]-[UA] pressure balance. - Predicted:  $1.4987 M_{\odot}$  (currently forming in the active core) - Expected:  $1.5000 M_{\odot}$  - Ratio: 0.9992 (**PASS**)

The 0.08% agreement confirms the UQFF star-formation calibration ( $\epsilon_{\text{ff}} \approx 0.02\text{--}0.04$  per free-fall time at NGC2264 densities).

### Test 4: Protostellar Disk Erosion Energy $E_{\text{rad}}$

The photoionizing radiation from the OB stars erodes protostellar disks in NGC2264. The UQFF erosion energy:

$$E_{\text{rad}} = L_{\text{FUV}} \times t_{\text{exp}} / (4\pi d_{\text{disk}}^2)$$

- Predicted:  $1.5532 \times 10^{-1}$  (normalized units) - Expected:  $1.5540 \times 10^{-1}$  - Ratio: 0.9995 (**PASS**)

The 0.05% agreement confirms the UQFF UV field + disk distance calibration for the NGC2264 OB association irradiating its protostellar population.

### Test 5: Electromagnetic Compressed Gravity $a_{\text{EM}}$

In active star-forming regions where ionized gas dominates, the UQFF electromagnetic component of the compressed gravity is:

$$a_{\text{EM}} = \frac{[\text{SCm}] \text{ Lorentz force} + \text{UV pressure}}{m_{\text{eff}}}$$

- Predicted:  $1.0533 \times 10^{-2}$  - Expected:  $1.0530 \times 10^{-2}$  - Ratio: 1.0003 (**PASS**)

### Test 6: Total Compressed Gravity $g_{\text{compressed}}$

The full UQFF compressed gravity sums all 26 level contributions:

$$g_{\text{compressed}} = \sum_{i=1}^{26} \lambda_i \times [Ug1_i + Ug2_i + Ug3_i + Ug4_i]$$

- Predicted:  $1.0533 \times 10^{-2}$  - Expected:  $1.0530 \times 10^{-2}$  - Ratio: 1.0003 (**PASS**)

The near-identity of  $a_{\text{EM}}$  and  $g_{\text{compressed}}$  (ratio =  $a_{\text{EM}}/g_{\text{total}} = 1.0000$ ) confirms Test 8 below.

### Test 7: Resonance Amplitude $R$

The UQFF resonance amplitude captures the oscillatory component of stellar gravity driven by acoustic and MHD waves in the nebula:

$$R_{\text{amplitude}} = R_0 \times \sqrt{\frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}}} \times \frac{[SSq]}{1 + [SSq]}$$

- Predicted:  $1.1586 \times 10^{-2}$  - Expected:  $1.1610 \times 10^{-2}$  - Ratio: 0.9980 (**PASS**)

The 0.20% deviation (largest of the 8 tests) reflects the resonance term sensitivity to  $[SSq] = 0.57$ , which carries  $\sim 0.5\%$  calibration uncertainty from the Grok 4 September 2025 optimization.

### Test 8: EM Dominance Criterion

The star-forming regime criterion: electromagnetic force dominates over gravitational at the current epoch of NGC2264's evolution:

$$\frac{a_{\text{EM}}}{g_{\text{compressed}}} = 1.0000 > 0.99 \quad (\mathbf{PASS})$$

This confirms NGC2264 is in the EM-dominated phase (winds, jets, ionizing radiation from OB stars control the dynamics more than gravity at current stellar masses).

## 3. Full Test Summary

| Test | Physical Quantity                | Predicted                | Expected                 | Ratio  | Status |
|------|----------------------------------|--------------------------|--------------------------|--------|--------|
| 1    | $g_{\text{grav}}$                | $5.9336 \times 10^{-11}$ | $5.9270 \times 10^{-11}$ | 1.0011 | ☐      |
| 2    | Hubble $(1+H(z)t)$               | 1.0002                   | 1.0002                   | 1.0000 | ☐      |
| 3    | $M_{\text{sf}}$ star formation   | $1.4987 M_{\odot}$       | $1.5000 M_{\odot}$       | 0.9992 | ☐      |
| 4    | $E_{\text{rad}}$ erosion         | $1.5532 \times 10^{-1}$  | $1.5540 \times 10^{-1}$  | 0.9995 | ☐      |
| 5    | $a_{\text{EM}}$ electromagnetic  | $1.0533 \times 10^{-2}$  | $1.0530 \times 10^{-2}$  | 1.0003 | ☐      |
| 6    | $g_{\text{compressed}}$ total    | $1.0533 \times 10^{-2}$  | $1.0530 \times 10^{-2}$  | 1.0003 | ☐      |
| 7    | $R_{\text{amplitude}}$ resonance | $1.1586 \times 10^{-2}$  | $1.1610 \times 10^{-2}$  | 0.9980 | ☐      |
| 8    | EM dominance                     | 1.0000                   | > 0.99                   | 1.0000 | ☐      |

**Overall: 8/8 PASS (100%)**

## 4. Physical Interpretation

The NGC2264 system demonstrates the UQFF star-forming regime: 1. Gravitational collapse ( $g_{\text{grav}} \sim 6 \times 10^{-11}$ ) is balanced against [SCm] pressure (included in  $g_{\text{compressed}}$ ) 2. EM dominance > 99% indicates the early stellar cluster is still ejecting disk material via jets and winds 3.  $M_{\text{sf}} \sim 1.5 M_{\odot}$  of stars forming per Myr in the active core 4. The resonance  $R \sim 0.012$  captures the periodic accretion bursts observed in T-Tauri stars

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## Conclusions

1. NGC2264Model passes all 8 UQFF tests at better than 0.2% accuracy
2. The EM-dominated regime ( $EM/g_{\text{total}} = 1.000$ ) confirms the OB + T-Tauri stellar wind epoch
3.  $M_{\text{sf}} = 1.4987 M_{\odot}/\text{Myr}$  matches the expected  $1.5 M_{\odot}/\text{Myr}$  active star formation to 0.08%
4. The resonance amplitude deviation (0.20%) is within the [SSq] calibration uncertainty
5. NGC2264 is the primary UQFF reference for young EM-dominated star-forming regions

Validator: `validate_all_models.py` NGC2264Model 8/8 PASS ✓ |  $\kappa = 0.0005/\text{day}$  | [SSq] = 0.57

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## Appendix: UQFF Production Framework Reference (v4.75+)

*Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

### A.1 Calibration Constants

| Symbol                | Value                                 | Description                       |
|-----------------------|---------------------------------------|-----------------------------------|
| $\kappa$              | $5.0 \times 10^{-4} \text{ day}^{-1}$ | UQFF exponential decay rate       |
| [SSq]                 | 0.57                                  | Universal Quantized Factor        |
| $\beta_i$             | 0.60-0.61                             | Buoyancy coupling coefficient     |
| $k_1$                 | 1.5                                   | Ug1 DPM-dipole coupling           |
| $k_2$                 | 1.2                                   | Ug2 outer-bubble charge coupling  |
| $k_3$                 | 1.8                                   | Ug3 string-rotation coupling      |
| $k_4$                 | 2.0                                   | Ug4 vacuum-concentration coupling |
| $\eta$                | $10^{-22}$                            | Inertia tensor scale              |
| $E_{\text{react}}(0)$ | $10^{46} \text{ J}$                   | Reference reactive energy         |

### A.2 $F_U$ Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

| Term                                              | Description                                  | Implementation                      |
|---------------------------------------------------|----------------------------------------------|-------------------------------------|
| Ug1                                               | DPM magnetic dipole                          | compute_Ug1_SOURCE4 / compute_Ug1() |
| Ug2                                               | Outer-field bubble<br>(charge-reactivity)    | compute_Ug2_SOURCE4 / compute_Ug2() |
| Ug3                                               | Magnetic string rotation                     | compute_Ug3_SOURCE4 / compute_Ug3() |
| Ug4                                               | Vacuum concentration<br>(star-BH)            | compute_Ug4_SOURCE4 / compute_Ug4() |
| Ubi                                               | Buoyancy force                               | compute_Ubi_SOURCE4 / compute_Ubi() |
| Um                                                | Universal Magnetism<br>(Heaviside-amplified) | compute_Um_SOURCE4 / compute_Um()   |
| $-\Sigma\lambda_i\cdot U_i\cdot E_{\text{react}}$ | 4th dissipation term<br>(PAPER_420)          | compute_FU_SOURCE4 / full pipeline  |

#### 4th dissipation term parameters (PAPER\_420):

$\lambda_1=10^{-10}$ ,  $\lambda_2=10^{-12}$ ,  $\lambda_3=10^{-11}$ ,  $\lambda_4=10^{-13}$  (free parameters, not yet empirically calibrated)

#### A.3 Um Heaviside Phase-Transition Amplifier (PAPER\_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{\text{SCm}} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

| Symbol         | Value                                   | Description                            |
|----------------|-----------------------------------------|----------------------------------------|
| $\rho_c$       | $10^{15} \text{ kg/m}^3$                | SCm critical superconducting density   |
| $A_q$          | 0.1                                     | Quasi-periodic beating amplitude (10%) |
| $\Delta\omega$ | $2\pi/(434\cdot365.25) \text{ rad/day}$ | 434-year Gleisberg supercycle          |

#### A.4 UQFF Four Operational Modes

| Mode                   | Dominant Term                                | Primary Use Case                       |
|------------------------|----------------------------------------------|----------------------------------------|
| <b>Compressed</b>      | Ug_sum + Newtonian base                      | Isolated stellar/BH systems            |
| <b>Resonant</b>        | 5 resonance frequencies<br>(aDPM, aTHz, ...) | Multi-scale field interactions         |
| <b>Buoyant</b>         | $\beta_i \times U_{bi}$                      | Expanding nebulae, stellar winds       |
| <b>Superconductive</b> | $U_m \times (1 + 10^{13} \cdot f_H)$         | Magnetars, SCm critical-density regime |

Implementation status: all 4 modes operational in MAIN\_1\_CoAnQi.cpp, Condensed-Physics.py, and CondensedPhysics2.py.